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Accident Modeling, Risk Assessment, and Management

3.1 Introduction

Toxicology is defined as a qualitative and quantitative study of the adverse effects of toxicants on biological organisms. A toxicant can be a chemical or physical agent, including dusts, fibers, noise, and radiation. Toxicants enter the human body through any of the following: (i) ingestion, which is through the mouth into stomach; (ii) inhalation, which is through the mouth or nose into lungs; (iii) injection, which is through the cuts into skin; and (iv) dermal absorption, which is through the skin membrane.

3.2 Dose Versus Response

Biological organisms respond differently to the same dose of toxicants. Factors responsible for such variation in their response are age, sex, weight, diet, and general health conditions. Figure 3.1 shows the response behavior for the dose of the toxicants. Figure 3.2 shows the logarithmic plot of the dose–response behavior under the influence of toxicants.

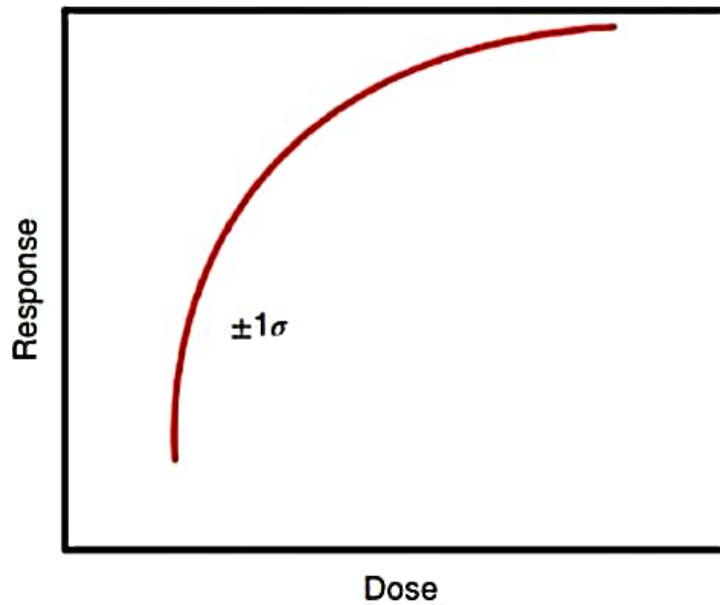


Figure 3.1 Dose vs. response behavior of toxicants

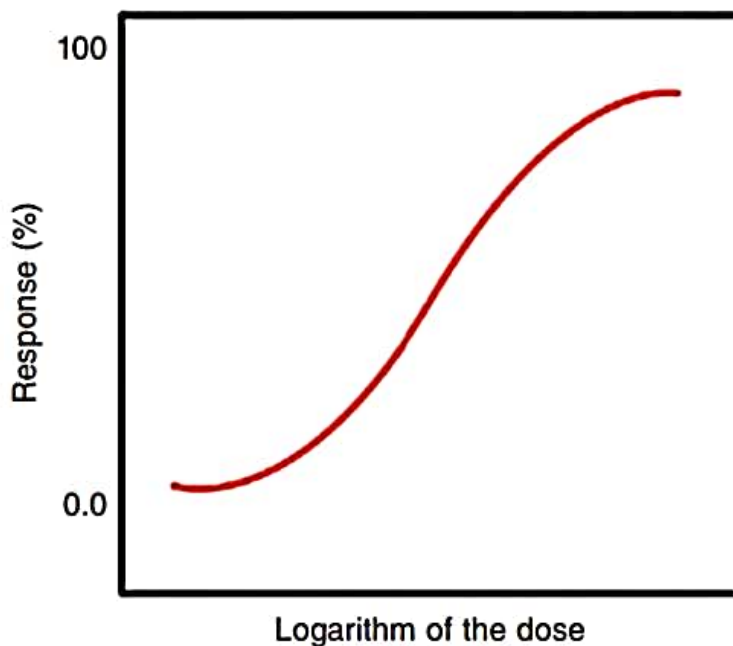


Figure 3.2 Logarithmic dose vs. response behavior of toxicants

3.2.1 Various Types of Doses

Different types of doses are briefly discussed next and the plot is shown in Figure 3.3.

(a) Lethal Dose (LD)

If the response to the chemical or agent is lethal and deadly, the response versus log dose curve is called LD curve.

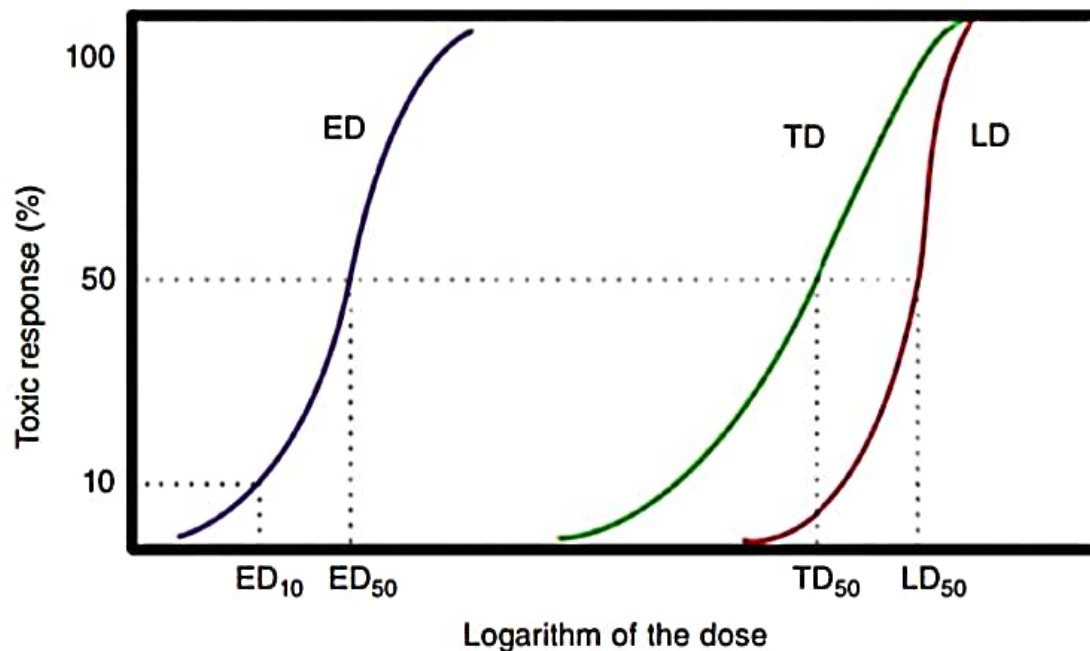


Figure 3.3 Various types of doses

(b) Effective Dose (ED)

If the response to the chemical or agent is minor and reversible (e.g., minor eye irritation, skin infection, sour throat, eye burning sensation etc.), the response versus log dose curve is called ED curve.

(c) Toxic Dose (TD)

If the response to the agent is toxic (i.e., it causes an undesirable response that is not lethal but irreversible, e.g., liver or lung damage), the response-log dose curve is called TD curve.

(d) Lethal Concentration (LC)

For gas concentration, logarithm of the dose is used.

3.2.2 Threshold Limit Value (TLV) Concentration

The TLV concentration is based on 760 Hg pressure at 25°C and a molar volume of 24.45 l. The equations for converting parts per million (ppm) to mg/m³ and vice versa are as follows:

$$\text{TLV in mg/m}^3 = \frac{(\text{TLV in ppm}) \times (\text{gram molecular weight of substance})}{24.45} \quad (3.1)$$

$$\text{TLV in ppm} = \frac{(\text{TLV in mg/m}^3) \times 24.45}{(\text{gram molecular weight of substance})} \quad (3.2)$$

3.29 Elements to Control Process Accidents

Chemical plants are potential source of accidents. Oil spills, fire, explosions, reactor runaway and fugitive emissions are the potential hazards that are responsible for process accidents. Mechanical hazards include improper maintenance, tripping, falling, or moving equipment. These are main contributors to cause spills, fire, and explosions in chemical industries. Some of the common types of chemical accidents are given in Table 3.7.

Hazardous materials

Flammable materials, combustible materials, toxic chemicals, unstable materials, highly reactive reactants are called as hazardous materials.

Initiating events

Equipment malfunctions, containment failures, thermal runaway, human error in operations, maintenance, etc. are examples of initiating events in any accident.

Intermediate propagating events

Process parameters like pressure, temperature, flow rate and their deviations, toxic materials, reactive materials, ignition/explosion are examples of intermediate propagating events.

Intermediate mitigating events

Safety system responses (e.g., relief valves, grounding, back-up utilities), mitigation system responses (e.g., vents, blow-out walls/ceilings, containment dikes, flares, sprinklers, etc.), contingency operations like alarms, emergency procedures, personnel safety equipment, evacuations, security are different intermediate mitigating events that can control process accidents.

Accident consequences

Fire, explosions, dispersion of toxic chemicals are the severe consequences that arise from process accidents.

- *Steps in Risk analysis for process accidents*
- Predict the accident occurrence and its damage potential
- Reduce the envisaged risk well in advance of an accident
- Ensure system safety during operations

3.3 Fire and Explosion Modeling

Chemical process systems contain substantial hazard that arise from fire and explosions (Ramamurthy, 2011). Three common chemical plant accidents are fire, explosion, and toxic releases (Chamberlain, 1987).

3.3.1 Fundamentals of Fire and Explosion

(a) Fire

Fire is a rapid exothermal oxidation of ignited fuel. Fuel can be in solid, liquid, or vapor form. Fire will release energy in the form of thermal radiation, which takes time to reach its peak intensity. Fire can also be formed as a result of explosion.

(b) Explosion

Explosion is a rapid expansion of gases resulting from (rapidly moving) pressure or shock waves. Expansion can be mechanical or resulting from a chemical reaction. Explosion damage is caused by the pressure or shock waves, which release energy rapidly that can cause fire (Chuan-Jie et al., 2013).

(c) Accident Prevention

Fire and explosion accidents can be prevented using the knowledge of fire and explosion characteristics of materials, nature of fire, and the explosion process. The procedure to reduce fire and explosion hazards is by using fire triangle as shown in Figure 3.4. Fire and explosion can be prevented by removing any one of the arms of the fire triangle.

3.4 Fire and Explosion Characteristics of Materials

Fire characteristics of flammable materials are as follows:

(a) Auto-ignition temperature (AIT)

It is the fixed temperature above which the material may not require any external ignition source for combustion.

(b) Flash point

It is the lowest temperature at which liquid gives up enough vapor to maintain continuous flame.

(c) Flammability limits

It is the range of vapor concentration that could cause combustion on meeting the ignition source. There are two limits in which the fuel will

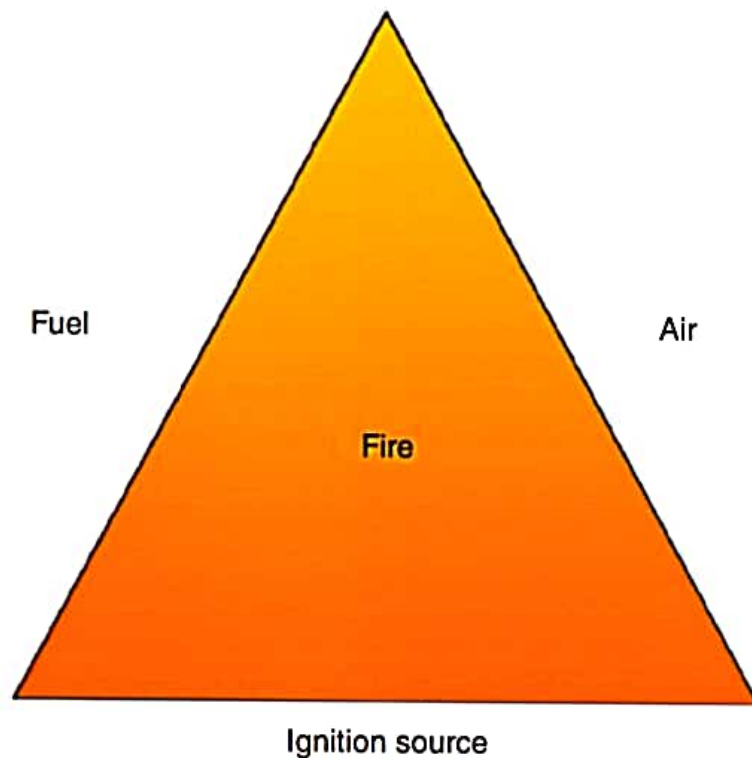


Figure 3.4 Fire triangle

catch fire: Lower Flammability Limit (LFL) and Upper Flammability Limit (UFL). LFL is the limit below which the mixture will not burn due to lean mixture and UFL is the limit above which the mixture will not catch fire as the mixture is too rich to catch fire.

(d) Limiting oxygen concentration (LOC)

It is the minimum oxygen concentration below which combustion is not possible, with any fuel mixture. It is expressed as volume percentage of oxygen. It is also called as Minimum Oxygen Concentration (MOC) or Maximum Safe Oxygen Concentration (MSOC).

(e) Shock wave

These are abrupt pressure waves moving through a medium. A shock wave in open air is usually followed by wind, which is called a blast wave. One of the important characteristics of the shock wave is that the pressure increase in shock wave is so rapid that the process is mostly adiabatic.

(f) Over pressure

It is pressure that is imparted on an object by a shock wave.

Figure 3.5 shows the flammability characteristics of materials.

3.7 Flammability Diagram for Hydrocarbons

Flammability diagram determines whether or not a given mixture is flammable. This will be helpful to control or prevent fire and explosion of flammable mixtures. Flammability diagram depends on the chemical species and is a function of temperature and pressure. A typical flammability diagram is shown in Figure 3.6.

3.7.1 Constructing Flammability Diagram

The construction of flammability diagram is done using the following steps:

- Mark three arms of the triangle and mark them as oxygen, nitrogen, and fuel arms respectively. Apex and origin of these arms are marked in an anti-clockwise order as shown in Figure 3.6.
- Draw the air line by connecting apex of the fuel arm with that of 79% of nitrogen. This is called as an air line.
- Draw LFL and UFL of the fuel mixture on the air line (% fuel in air).
- Locate the stoichiometric point on the oxygen axis.
- Draw a stoichiometric line from this point to 100% nitrogen apex.
- Locate LOC on the oxygen axis and draw a line parallel to the fuel arm until it intersects the stoichiometric line.
- Draw a point at this intersection.
- Draw LFL and UFL in pure oxygen, if known (% of fuel in pure oxygen).
- Connect the points to get the flammability diagram, as shown in the figure.

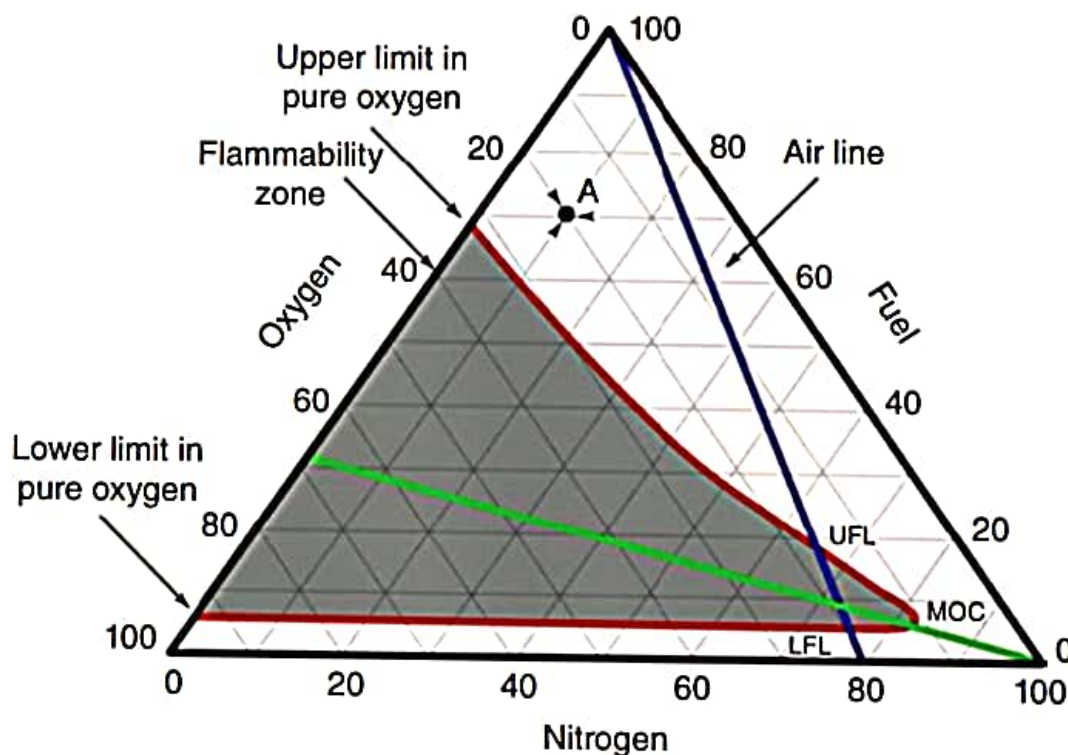


Figure 3.6 Flammability diagram

3.9 Explosions

Explosion is a rapid release of energy causing development of pressure or shock waves. There are different types of industrial explosions as discussed next:

Confined Vapor Cloud Explosion (CVCE):

It is a type of explosion that happens in a vessel or in a confined space (e.g., inside a building). It is generally caused due to the release of high pressure or chemical energy.

Vapor Cloud Explosion (VCE):

It is a type of explosion caused by the instantaneous vapor cloud, which is formed in air due to the release of flammable chemicals into atmosphere.

Boiling Liquid Expanding Vapor Explosion (BLEVE):

BLEVE is caused due to the instantaneous release of a large amount of vapor through narrow opening under the pressurized condition (Tasneem Abbasi and Abbasi, 2007).

Vented Explosion (VE)

It is caused due to venting of chemicals at a high velocity.

Dust Explosion

Dust explosion is caused due to the rapid combustion of fine solid particles.

3.10 Explosion Characteristics

Explosion energy is dissipated in different forms: (i) pressure wave; (ii) projectiles; (iii) thermal radiation; and (iv) acoustic energy (Planas-Cuchi et al., 2004). Some of the types of energy, which is generated due to explosion are discussed next:

Blast wave is a shock wave in open air, which is generally followed by a strong wind. *Overpressure* of an object is a result of an impacting shock waves on any object. *Detonation* is a kind of explosion in which reaction front moves at a speed greater than that of sound in the given medium. *Deflagration* is a kind of explosion in which reaction front (energy front) moves at a lesser speed than that of sound in the given medium.

3.11 Explosion Modeling

Explosions result in a blast or a pressure wave moving out from the explosion center at the speed of sound. Shock wave or overpressure is the basic cause for damages during explosions. Missiles or projectiles are other important

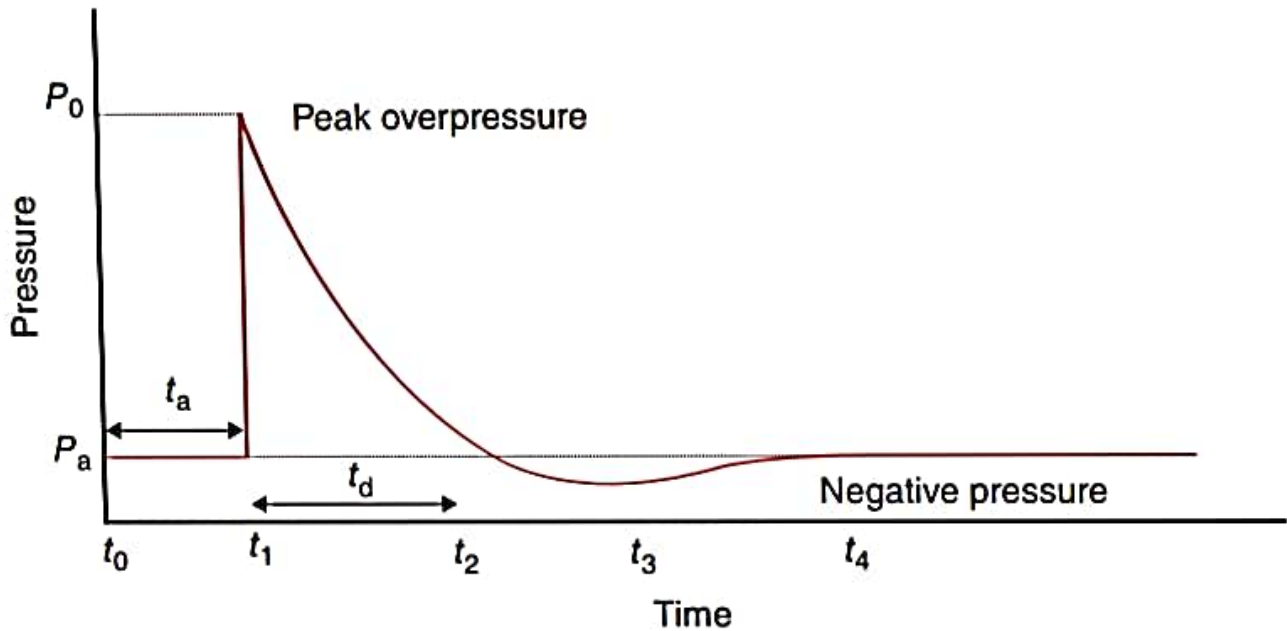


Figure 3.8 Variation of overpressure with time

sources of damage. Generally, damage caused by explosions is a function of rate of pressure increase and the duration of blast wave. As explosion generates a rapid rise in pressure, damage caused by blasts is estimated based on the peak side-on overpressure. The propagating wave causes damage to the objects located on its path, which is then followed by a negative pressure wave. This causes further damage before the pressure waves return to the atmospheric pressure. Therefore, damage depends upon various factors: (i) maximum pressure reached; (ii) velocity of propagation; and (iii) environmental characteristics. The variation of overpressure with time due to blast wave is given in Figure 3.8. The area under the curve is the measure of severity of explosion.

3.13 Energy in Chemical Explosions

Energy released during a reaction is computed using standard thermodynamics. The heat of combustion is used as a mode to assess the explosion strength. From the past studies, it can be seen that the explosion energy differs by about 10% from that of the value of heat of combustion.

3.14 Explosion Energy in Physical Explosions

In the mechanical or physical explosions, a reaction does not occur. Energy is obtained from the energy content of the contained substance. Four common approaches used are: (i) Brode's method; (ii) isentropic expansion method; (iii) isothermal expansion method; and (iv) laws of thermodynamics. Equations (3.22–3.24) are used in place of the (first) three methods:

$$E_{\text{Brode}} = \frac{(P_2 - P_1) V}{\gamma - 1} \quad (3.22)$$

$$E_{\text{isentropic}} = \frac{P_2 V}{\gamma - 1} \left[1 - \left[\frac{P_1}{P_2} \right]^{\frac{\gamma-1}{\gamma}} \right] \quad (3.23)$$

$$E_{\text{isothermal}} = P_2 V \ln \left(\frac{P_2}{P_1} \right) \quad (3.24)$$

where, E is the explosion energy; P_1 is the atmospheric pressure, P_2 is the bursting pressure, V is the volume of vessel, and γ is the heat capacity ratio.

3.15 Dust and Gaseous Explosion

Gas molecules are smaller and well-defined in size, whereas dust particles vary in size; this variation in magnitude is larger than molecules. K_G and K_{st} are deflagration index for gas and dust, respectively, which are given by Equations (3.25) and (3.26), respectively. Variation in pressure with respect to time for gas and dust explosion is given in Figure 3.10.

$$\left(\frac{dp}{dt} \right)_{\max} \sqrt[3]{V} = K_{st} \quad (3.25)$$

$$\left(\frac{dp}{dt} \right)_{\max} \sqrt[3]{V} = K_G \quad (3.26)$$

3.17 Fire and Explosion Preventive Measures

Some of the strategies that are commonly used in oil and gas industries are discussed in the next subsection:

3.17.1 Inerting and Purging

This procedure aims to reduce oxygen or fuel concentration lower than a target value using an inert gas. While nitrogen, carbon dioxide, and others are possible choice of inert gases, nitrogen is commonly used at a control point of 4% below LOC. Some of the purging methods are vacuum purging, pressure purging, combined purging, vacuum and pressure purging with impure nitrogen, sweep-through purging, and siphon purging. During purging methods, it is assumed that pure nitrogen is purged, which mixes well inside the vessel and therefore ideal gas behavior.

3.17.1.1 Vacuum Purging

It is one of the most commonly used inerting procedures for vessels. In vacuum purging, the vessel is evacuated and replaced with inert gas. The procedure includes drawing vacuum from the vessel and then replace it with inert gas. This cycle is repeated until the desired concentration is reached. The cycle is shown in Figure 3.12.

Initial oxidant concentration under vacuum (y_0) is the same as the initial concentration. Number of moles at initial high pressure (P_H) and initial low pressure (P_L) are computed using an initial equation of state, for an ideal gas behavior. n_H and n_L are total moles in the atmosphere and vacuum states, respectively. At point A in the figure, the number of oxidants is calculated using Equation (3.27) and that at point B, using Equation (3.28).

$$\text{at A, } n_{\text{oxygen}} = y_0 \left[\frac{P_L V}{R_g T} \right] \quad (3.27)$$

3.23 Toxic Release and Dispersion Modeling

3.23.1 Threshold Limit Values (TLVs)

TLVs represent conditions to which all workers will be repeatedly exposed every day without adverse health effects. For any dose value below this dosage, human body can be detoxified. The two agencies who established TLVs are The American Conference of Governmental Industrial Hygienists (ACGIH) and Occupational Safety and Health Administration, U.S. (OSHA) who defined the Permissible Exposure Levels (PELs) for personnel working in process industries. TLVs are of three types which are:

TLV-TWA (Time Weighted Average):

This is an average of normal working per day (8 hours) or 40 work hours per week on an average for which workers will be exposed.

TLV-STEL (Short-term Exposure Limit):

It is the maximum concentration to which workers can be exposed to for a period of up to 15 minutes continuously without suffering. The consequences are intolerable irritation, chronic or irreversible tissue changes, and narcosis of sufficient degree that reduces worker's efficiency considerably.

TLV-C (Ceiling Limit):

It is the concentration that should not be exceeded, even instantaneously or else it results in fatal death. The conversion of TLVs from ppm to mg/m³ is based on 760 mm Hg pressure at 25°C and a molar volume of 24.45 l, which is given by:

$$\text{TLV in mg/m}^3 = \frac{(\text{TLV in ppm}) \times (\text{gram molecular weight of substance})}{24.45} \quad (3.43)$$

$$\text{TLV in ppm} = \frac{(\text{TLV in mg / m}^3) \times 24.45}{(\text{gram molecular weight of substance})} \quad (3.44)$$